

BCA MATHEMATICS-I (BCA-0101)

Complete Unit-IV Study Material

As per HPU BCA Semester-I Syllabus

Contents

1	Functions	3
1.1	Definition	3
1.2	Types of Functions	3
1.3	Domain and Range of Common Functions	3
1.4	Composite Functions	3
1.5	Inverse Functions	3
2	Limits and Continuity	4
2.1	Definition of Limit	4
2.2	Left and Right Hand Limits	4
2.3	Standard Limits	4
2.4	Properties of Limits	4
2.5	Continuity	5
2.6	Types of Discontinuity	5
3	Derivative of Functions	6
3.1	Definition	6
3.2	Basic Differentiation Formulas	6
3.3	Chain Rule	6
3.4	Product Rule	6
3.5	Quotient Rule	6
4	Maxima and Minima	7
4.1	First Derivative Test	7
4.2	Second Derivative Test	7
4.3	Absolute Maxima/Minima on $[a, b]$	7
5	Indefinite Integrals	8
5.1	Definition	8
5.2	Basic Integration Formulas	8
5.3	Properties	8
5.4	Substitution Method	8
5.5	Integration by Parts	8

6	Definite Integrals	10
6.1	Definition	10
6.2	Fundamental Theorem of Calculus	10
6.3	Properties	10
6.4	Even and Odd Functions	10
6.5	Area Under Curves	10
7	Practice Questions with Solutions	11
7.1	Functions (10 Questions)	11
7.2	Limits (10 Questions)	12
7.3	Continuity (5 Questions)	13
7.4	Derivatives (10 Questions)	14
7.5	Maxima and Minima (10 Questions)	15
7.6	Indefinite Integrals (10 Questions)	17
7.7	Definite Integrals (10 Questions)	18
8	HPU Exam Important Questions	19
8.1	Section-A (5 Marks)	19
8.2	Section-B (5 Marks)	19
8.3	Section-C (10 Marks)	19
8.4	Section-E (1 Mark)	19
9	Quick Revision Formulas	21
9.1	Functions	21
9.2	Limits	21
9.3	Derivatives	21
9.4	Integrals	21
9.5	Definite Integrals	21
9.6	Maxima/Minima	22

1 Functions

1.1 Definition

A function is a relation where each input has exactly one output.

$$f : A \rightarrow B$$

Domain: Set of all input values (x)

Co-domain: Set of all possible outputs

Range: Set of actual outputs

1.2 Types of Functions

Type	Definition	Example
One-One (Injective)	Unique output for each input	$f(x) = 2x$
Onto (Surjective)	Every output is covered	$f(x) = x + 1$
Bijjective	One-One + Onto	$f(x) = x$
Constant	Same output for all	$f(x) = 5$
Linear	$f(x) = mx + c$	$f(x) = 2x + 3$
Quadratic	$f(x) = ax^2 + bx + c$	$f(x) = x^2$

1.3 Domain and Range of Common Functions

Function	Domain	Range
$f(x) = x^2$	All real	$[0, \infty)$
$f(x) = 1/x$	$x \neq 0$	$y \neq 0$
$f(x) = \sqrt{x}$	$[0, \infty)$	$[0, \infty)$
$f(x) = \sin x$	All real	$[-1, 1]$
$f(x) = \cos x$	All real	$[-1, 1]$
$f(x) = \tan x$	$x \neq (2n + 1)\pi/2$	All real
$f(x) = \log_a x$	$(0, \infty)$	All real
$f(x) = a^x$	All real	$(0, \infty)$

1.4 Composite Functions

$$(g \circ f)(x) = g(f(x))$$

Example: $f(x) = 2x$, $g(x) = x + 3$

$$(g \circ f)(x) = g(2x) = 2x + 3$$

$$(f \circ g)(x) = f(x + 3) = 2x + 6$$

1.5 Inverse Functions

If f is bijective:

$$f^{-1}(f(x)) = x, \quad f(f^{-1}(x)) = x$$

2 Limits and Continuity

2.1 Definition of Limit

$$\lim_{x \rightarrow a} f(x) = L$$

2.2 Left and Right Hand Limits

$$\lim_{x \rightarrow a^-} f(x) = L_1 \quad (\text{LHL})$$

$$\lim_{x \rightarrow a^+} f(x) = L_2 \quad (\text{RHL})$$

Limit exists if:

$$\text{LHL} = \text{RHL} = L$$

2.3 Standard Limits

Limit	Result
$\lim_{x \rightarrow 0} \frac{\sin x}{x}$	1
$\lim_{x \rightarrow 0} \frac{\tan x}{x}$	1
$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2}$	1/2
$\lim_{x \rightarrow 0} \frac{e^x - 1}{x}$	1
$\lim_{x \rightarrow 0} \frac{a^x - 1}{x}$	$\ln a$
$\lim_{x \rightarrow 0} \frac{\ln(1+x)}{x}$	1
$\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x$	e
$\lim_{x \rightarrow 0} (1+x)^{1/x}$	e

2.4 Properties of Limits

If $\lim f(x) = L$ and $\lim g(x) = M$:

1. $\lim[f(x) \pm g(x)] = L \pm M$
2. $\lim[cf(x)] = cL$
3. $\lim[f(x)g(x)] = LM$
4. $\lim \frac{f(x)}{g(x)} = \frac{L}{M}, M \neq 0$

2.5 Continuity

A function is continuous at $x = a$ if:

$$\lim_{x \rightarrow a} f(x) = f(a)$$

Conditions:

1. $f(a)$ exists
2. $\lim_{x \rightarrow a} f(x)$ exists
3. $\lim_{x \rightarrow a} f(x) = f(a)$

2.6 Types of Discontinuity

Type	Example
Removable	$f(x) = \frac{x^2-1}{x-1}$ at $x = 1$
Jump	$f(x) = \frac{ x }{x}$ at $x = 0$
Infinite	$f(x) = \frac{1}{x}$ at $x = 0$

3 Derivative of Functions

3.1 Definition

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

3.2 Basic Differentiation Formulas

Function	Derivative
$\frac{d}{dx}(c)$	0
$\frac{d}{dx}(x^n)$	nx^{n-1}
$\frac{d}{dx}(e^x)$	e^x
$\frac{d}{dx}(a^x)$	$a^x \ln a$
$\frac{d}{dx}(\ln x)$	$1/x$
$\frac{d}{dx}(\sin x)$	$\cos x$
$\frac{d}{dx}(\cos x)$	$-\sin x$
$\frac{d}{dx}(\tan x)$	$\sec^2 x$
$\frac{d}{dx}(\cot x)$	$-\csc^2 x$
$\frac{d}{dx}(\sec x)$	$\sec x \tan x$
$\frac{d}{dx}(\csc x)$	$-\csc x \cot x$

3.3 Chain Rule

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

Example: $y = \sin(3x)$, $\frac{dy}{dx} = 3 \cos(3x)$

3.4 Product Rule

$$\frac{d}{dx}(uv) = u \frac{dv}{dx} + v \frac{du}{dx}$$

3.5 Quotient Rule

$$\frac{d}{dx} \left(\frac{u}{v} \right) = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

4 Maxima and Minima

4.1 First Derivative Test

1. Find $f'(x)$
2. Set $f'(x) = 0 \rightarrow$ Critical points
3. Check sign change:
 - $+$ to $- \rightarrow$ Maximum
 - $-$ to $+$ $\rightarrow$ Minimum
 - No change $\rightarrow$ Point of Inflection

4.2 Second Derivative Test

1. Find $f'(x)$ and $f''(x)$
2. Set $f'(x) = 0 \rightarrow$ Critical points
3. Check $f''(x)$:
 - $f''(x) > 0 \rightarrow$ Minimum
 - $f''(x) < 0 \rightarrow$ Maximum
 - $f''(x) = 0 \rightarrow$ Inconclusive

4.3 Absolute Maxima/Minima on $[a, b]$

1. Find critical points in (a, b)
2. Evaluate $f(a)$, $f(b)$, $f(c)$ for all critical points
3. Largest = Absolute Maximum
4. Smallest = Absolute Minimum

5 Indefinite Integrals

5.1 Definition

$$\int f(x) dx = F(x) + C$$

where $F'(x) = f(x)$

5.2 Basic Integration Formulas

Function	Integral
$\int x^n dx$	$\frac{x^{n+1}}{n+1} + C, n \neq -1$
$\int \frac{1}{x} dx$	$\ln x + C$
$\int e^x dx$	$e^x + C$
$\int a^x dx$	$\frac{a^x}{\ln a} + C$
$\int \sin x dx$	$-\cos x + C$
$\int \cos x dx$	$\sin x + C$
$\int \sec^2 x dx$	$\tan x + C$
$\int \csc^2 x dx$	$-\cot x + C$
$\int \sec x \tan x dx$	$\sec x + C$
$\int \frac{1}{\sqrt{1-x^2}} dx$	$\sin^{-1} x + C$
$\int \frac{1}{1+x^2} dx$	$\tan^{-1} x + C$

5.3 Properties

- $\int [f(x) \pm g(x)] dx = \int f(x) dx \pm \int g(x) dx$
- $\int k f(x) dx = k \int f(x) dx$

5.4 Substitution Method

Let $u = g(x)$, $du = g'(x) dx$:

$$\int f(g(x))g'(x) dx = \int f(u) du$$

5.5 Integration by Parts

$$\int u dv = uv - \int v du$$

ILATE Rule: I-L-A-T-E

- I = Inverse trig
- L = Logarithmic
- A = Algebraic
- T = Trigonometric
- E = Exponential

6 Definite Integrals

6.1 Definition

$$\int_a^b f(x) dx = [F(x)]_a^b = F(b) - F(a)$$

6.2 Fundamental Theorem of Calculus

$$\frac{d}{dx} \int_a^x f(t) dt = f(x)$$

$$\int_a^b f(x) dx = F(b) - F(a)$$

6.3 Properties

1. $\int_a^b f(x) dx = - \int_b^a f(x) dx$
2. $\int_a^a f(x) dx = 0$
3. $\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$
4. $\int_a^b f(x) dx = \int_a^b f(a+b-x) dx$
5. $\int_{-a}^a f(x) dx = \begin{cases} 2 \int_0^a f(x) dx, & \text{even} \\ 0, & \text{odd} \end{cases}$
6. $\int_0^a f(x) dx = \int_0^a f(a-x) dx$

6.4 Even and Odd Functions

- **Even:** $f(-x) = f(x)$
- **Odd:** $f(-x) = -f(x)$

6.5 Area Under Curves

Area between $y = f(x)$, x-axis, $x = a$, $x = b$:

$$A = \int_a^b f(x) dx$$

Area between $y = f(x)$ and $y = g(x)$:

$$A = \int_a^b |f(x) - g(x)| dx$$

7 Practice Questions with Solutions

7.1 Functions (10 Questions)

Q1. Find domain of $f(x) = \sqrt{x-2}$.

Solution: $x-2 \geq 0 \rightarrow x \geq 2 \rightarrow \text{Domain} = [2, \infty)$

Q2. If $f(x) = 2x + 3$ and $g(x) = x^2$, find $(f \circ g)(x)$.

Solution: $(f \circ g)(x) = f(x^2) = 2x^2 + 3$

Q3. Find domain of $f(x) = \frac{1}{x^2-4}$.

Solution: $x^2 - 4 \neq 0 \rightarrow x \neq \pm 2 \rightarrow \text{Domain} = \mathbb{R} - \{-2, 2\}$

Q4. Find $f^{-1}(x)$ for $f(x) = 3x + 2$.

Solution: $y = 3x + 2 \rightarrow x = (y - 2)/3 \rightarrow f^{-1}(x) = (x - 2)/3$

Q5. If $f(x) = x^2 + 2x$, find $f(2)$ and $f(-2)$.

Solution: $f(2) = 4 + 4 = 8$, $f(-2) = 4 - 4 = 0$

Q6. Find the range of $f(x) = \sin x$ on $[0, \pi]$.

Solution: Range = $[0, 1]$

Q7. If $f(x) = x^2$ and $g(x) = 2x$, find $(f \circ g)(x)$.

Solution: $(f \circ g)(x) = f(2x) = (2x)^2 = 4x^2$

Q8. Find the domain of $f(x) = \log(x - 3)$.

Solution: $x - 3 > 0 \rightarrow x > 3 \rightarrow \text{Domain} = (3, \infty)$

Q9. If $f(x) = \frac{x+1}{x-1}$, find $f(2)$.

Solution: $f(2) = \frac{2+1}{2-1} = 3$

Q10. Find the range of $f(x) = x^2 + 1$.

Solution: Range = $[1, \infty)$

7.2 Limits (10 Questions)

Q11. Evaluate: $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$

Solution: $\lim_{x \rightarrow 2} \frac{(x-2)(x+2)}{x-2} = \lim_{x \rightarrow 2} (x+2) = 4$

Q12. Evaluate: $\lim_{x \rightarrow 0} \frac{\sin 2x}{x}$

Solution: $\lim_{x \rightarrow 0} \frac{\sin 2x}{2x} \times 2 = 1 \times 2 = 2$

Q13. Evaluate: $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2}$

Solution: $\frac{1}{2}$

Q14. Evaluate: $\lim_{x \rightarrow \infty} \frac{2x^2 + 3x + 1}{x^2 - 5x + 4}$

Solution: Divide by x^2 : $\lim_{x \rightarrow \infty} \frac{2 + 3/x + 1/x^2}{1 - 5/x + 4/x^2} = 2$

Q15. Evaluate: $\lim_{x \rightarrow 0} \frac{e^{2x} - 1}{x}$

Solution: $\lim_{x \rightarrow 0} \frac{e^{2x} - 1}{2x} \times 2 = 2$

Q16. Evaluate: $\lim_{x \rightarrow 0} \frac{\tan 3x}{x}$

Solution: $\lim_{x \rightarrow 0} \frac{\tan 3x}{3x} \times 3 = 1 \times 3 = 3$

Q17. Evaluate: $\lim_{x \rightarrow 0} \frac{\ln(1+x)}{x}$

Solution: 1 (Standard limit)

Q18. Evaluate: $\lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1}$

Solution: $\lim_{x \rightarrow 1} \frac{(x-1)(x+1)}{x-1} = \lim_{x \rightarrow 1} (x+1) = 2$

Q19. Evaluate: $\lim_{x \rightarrow 0} \frac{\sin^2 x}{x^2}$

Solution: $\left(\lim_{x \rightarrow 0} \frac{\sin x}{x} \right)^2 = 1$

Q20. Evaluate: $\lim_{x \rightarrow \infty} \frac{3x + 1}{x + 2}$

Solution: Divide by x : $\lim_{x \rightarrow \infty} \frac{3 + 1/x}{1 + 2/x} = 3$

7.3 Continuity (5 Questions)

Q21. Check continuity of $f(x) = x^2$ at $x = 2$.

Solution: $f(2) = 4$, $\lim_{x \rightarrow 2} x^2 = 4$, $\lim f(x) = f(2) \rightarrow$ Continuous

Q22. Find k so that

$$f(x) = \begin{cases} kx + 5, & x \leq 2 \\ x - 1, & x > 2 \end{cases}$$

is continuous.

Solution: LHL = $2k + 5$, RHL = 1

$2k + 5 = 1 \rightarrow 2k = -4 \rightarrow k = -2$

Q23. Check continuity of $f(x) = \frac{x^2-4}{x-2}$ at $x = 2$.

Solution: $f(2)$ is undefined, so not continuous.

Q24. Check continuity of $f(x) = |x|$ at $x = 0$.

Solution: LHL = RHL = 0, $f(0) = 0 \rightarrow$ Continuous

Q25. Find k so that $f(x) = \{kx^2, x < 2; 4x, x \geq 2\}$ is continuous.

Solution: LHL = $4k$, RHL = 8, $4k = 8 \rightarrow k = 2$

7.4 Derivatives (10 Questions)

Q26. Differentiate: $f(x) = x^5 + 3x^3 - 2x + 7$

Solution: $f'(x) = 5x^4 + 9x^2 - 2$

Q27. Differentiate: $f(x) = \sin(2x^2)$

Solution: $f'(x) = 4x \cos(2x^2)$

Q28. Differentiate: $f(x) = x^2 \sin x$

Solution: $f'(x) = x^2 \cos x + 2x \sin x$

Q29. Differentiate: $f(x) = e^{3x} \cos 2x$

Solution: $f'(x) = e^{3x}(3 \cos 2x - 2 \sin 2x)$

Q30. Differentiate: $f(x) = \ln(x^2 + 1)$

Solution: $f'(x) = \frac{2x}{x^2+1}$

Q31. Differentiate: $f(x) = \frac{x^2}{x+1}$

Solution: $f'(x) = \frac{(x+1)(2x) - x^2(1)}{(x+1)^2} = \frac{x^2+2x}{(x+1)^2}$

Q32. Differentiate: $f(x) = \tan(3x)$

Solution: $f'(x) = 3 \sec^2(3x)$

Q33. Differentiate: $f(x) = e^x \sin x$

Solution: $f'(x) = e^x(\sin x + \cos x)$

Q34. Differentiate: $f(x) = \sqrt{x^2 + 1}$

Solution: $f'(x) = \frac{x}{\sqrt{x^2+1}}$

Q35. Differentiate: $f(x) = \cos^3 x$

Solution: $f'(x) = -3 \cos^2 x \sin x$

7.5 Maxima and Minima (10 Questions)

Q36. Find max/min of $f(x) = x^2 - 4x + 5$.

Solution: $f'(x) = 2x - 4 = 0 \rightarrow x = 2$

$f''(x) = 2 > 0 \rightarrow$ Minimum

$f(2) = 4 - 8 + 5 = 1$ (Minimum)

Q37. Find max/min of $f(x) = x^3 - 3x^2 + 3$.

Solution: $f'(x) = 3x^2 - 6x = 3x(x - 2) = 0 \rightarrow x = 0, 2$

$f''(x) = 6x - 6$

$f''(0) = -6 < 0 \rightarrow$ Maximum

$f''(2) = 6 > 0 \rightarrow$ Minimum

$f(0) = 3$ (Maximum), $f(2) = -1$ (Minimum)

Q38. Find max/min of $f(x) = x^3 - 3x$.

Solution: $f'(x) = 3x^2 - 3 = 0 \rightarrow x = \pm 1$

$f''(x) = 6x$

$f''(1) = 6 > 0 \rightarrow$ Minimum, $f(1) = -2$

$f''(-1) = -6 < 0 \rightarrow$ Maximum, $f(-1) = 2$

Q39. Find max/min of $f(x) = 2x^3 - 3x^2 - 12x + 5$.

Solution: $f'(x) = 6x^2 - 6x - 12 = 6(x - 2)(x + 1) = 0 \rightarrow x = 2, -1$

$f''(x) = 12x - 6$

$f''(2) = 18 > 0 \rightarrow$ Minimum, $f(2) = 16 - 12 - 24 + 5 = -15$

$f''(-1) = -18 < 0 \rightarrow$ Maximum, $f(-1) = -2 - 3 + 12 + 5 = 12$

Q40. Find max/min of $f(x) = x + \frac{1}{x}$, $x > 0$.

Solution: $f'(x) = 1 - \frac{1}{x^2} = 0 \rightarrow x = 1$

$f''(x) = \frac{2}{x^3} > 0 \rightarrow$ Minimum

$f(1) = 2$ (Minimum)

Q41. Find max/min of $f(x) = \sin x$, $x \in [0, \pi]$.

Solution: $f'(x) = \cos x = 0 \rightarrow x = \pi/2$

$f''(x) = -\sin x$

$f''(\pi/2) = -1 < 0 \rightarrow$ Maximum

$f(\pi/2) = 1$ (Maximum), $f(0) = 0$, $f(\pi) = 0$ (Minima)

Q42. Find max/min of $f(x) = x^2 - 2x + 3$ on $[0, 3]$.

Solution: $f'(x) = 2x - 2 = 0 \rightarrow x = 1$

$f(0) = 3$, $f(1) = 2$ (Minimum), $f(3) = 6$ (Maximum)

Q43. Find max/min of $f(x) = x^3$ on $[-2, 2]$.

Solution: $f'(x) = 3x^2 = 0 \rightarrow x = 0$

$f(-2) = -8$ (Minimum), $f(0) = 0$, $f(2) = 8$ (Maximum)

Q44. Find the maximum value of $f(x) = -x^2 + 4x - 3$.

Solution: $f'(x) = -2x + 4 = 0 \rightarrow x = 2$

$f''(x) = -2 < 0 \rightarrow$ Maximum

$f(2) = -4 + 8 - 3 = 1$

Q45. Find max/min of $f(x) = x^4 - 2x^2$.

Solution: $f'(x) = 4x^3 - 4x = 4x(x - 1)(x + 1) = 0 \rightarrow x = 0, \pm 1$

$f''(x) = 12x^2 - 4$

$f''(1) = 8 > 0 \rightarrow$ Min, $f(1) = -1$

$$f''(-1) = 8 > 0 \rightarrow \text{Min}, f(-1) = -1$$

$$f''(0) = -4 < 0 \rightarrow \text{Max}, f(0) = 0$$

7.6 Indefinite Integrals (10 Questions)

Q46. $\int (x^4 + 2x^2 - 3x + 5) dx$

Solution: $\frac{x^5}{5} + \frac{2x^3}{3} - \frac{3x^2}{2} + 5x + C$

Q47. $\int \sin(3x + 2) dx$

Solution: $-\frac{\cos(3x+2)}{3} + C$

Q48. $\int xe^{x^2} dx$

Solution: Let $u = x^2$, $du = 2x dx$
 $= \frac{1}{2} \int e^u du = \frac{1}{2} e^{x^2} + C$

Q49. $\int \frac{2x}{x^2 + 1} dx$

Solution: Let $u = x^2 + 1$, $du = 2x dx$
 $= \int \frac{1}{u} du = \ln|x^2 + 1| + C$

Q50. $\int x \sin x dx$ (By Parts)

Solution: $u = x$, $dv = \sin x dx$, $du = dx$, $v = -\cos x$
 $= -x \cos x + \int \cos x dx = -x \cos x + \sin x + C$

Q51. $\int x^2 \cos x dx$ (By Parts)

Solution: $u = x^2$, $dv = \cos x dx$
 $= x^2 \sin x - 2 \int x \sin x dx$
 $= x^2 \sin x + 2x \cos x - 2 \sin x + C$

Q52. $\int e^x \sin x dx$ (By Parts twice)

Solution: $= \frac{e^x(\sin x - \cos x)}{2} + C$

Q53. $\int \frac{1}{x^2 + 4} dx$

Solution: $\frac{1}{2} \tan^{-1} \left(\frac{x}{2} \right) + C$

Q54. $\int \frac{x}{\sqrt{x^2 + 1}} dx$

Solution: $\sqrt{x^2 + 1} + C$

Q55. $\int \sec^2 x \tan x dx$

Solution: Let $u = \tan x$, $du = \sec^2 x dx$
 $= \int u du = \frac{\tan^2 x}{2} + C$

7.7 Definite Integrals (10 Questions)

Q56. $\int_0^1 x^2 dx$

Solution: $\left[\frac{x^3}{3}\right]_0^1 = \frac{1}{3}$

Q57. $\int_0^{\pi/2} \sin x dx$

Solution: $[-\cos x]_0^{\pi/2} = 0 + 1 = 1$

Q58. $\int_0^1 (2x + 1) dx$

Solution: $[x^2 + x]_0^1 = 2$

Q59. $\int_{-1}^1 x^3 dx$

Solution: Odd function $\rightarrow 0$

Q60. $\int_0^{\pi/2} \cos x dx$

Solution: $[\sin x]_0^{\pi/2} = 1$

Q61. $\int_0^1 e^x dx$

Solution: $[e^x]_0^1 = e - 1$

Q62. $\int_0^2 (x^2 + 1) dx$

Solution: $\left[\frac{x^3}{3} + x\right]_0^2 = \frac{8}{3} + 2 = \frac{14}{3}$

Q63. $\int_0^{\pi/2} \sin^2 x dx$

Solution: $\frac{\pi}{4}$

Q64. $\int_0^{\pi/2} \cos^2 x dx$

Solution: $\frac{\pi}{4}$

Q65. $\int_0^1 \frac{1}{x+1} dx$

Solution: $[\ln(x+1)]_0^1 = \ln 2$

8 HPU Exam Important Questions

8.1 Section-A (5 Marks)

Q1. Find domain and range of $f(x) = \sqrt{x-3}$.

Q2. Evaluate: $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$.

Q3. Differentiate: $f(x) = \sin(2x^2)$.

Q4. Find max/min of $f(x) = x^2 - 6x + 9$.

Q5. Evaluate: $\int (3x^2 + 2x + 1) dx$.

Q6. Evaluate: $\int_0^1 x^3 dx$.

Q7. Differentiate: $f(x) = e^{2x} \sin 3x$.

Q8. Check continuity of $f(x) = 2x + 1$ at $x = 1$.

8.2 Section-B (5 Marks)

Q9. If $f(x) = 2x + 5$ and $g(x) = x^2$, find $(f \circ g)(x)$.

Q10. Evaluate: $\lim_{x \rightarrow 0} \frac{\sin 3x}{x}$.

Q11. Differentiate: $f(x) = \ln(x^2 + 3x)$.

Q12. Find max/min of $f(x) = x^3 - 3x^2 + 2$.

Q13. Evaluate: $\int x e^x dx$.

Q14. Evaluate: $\int_0^{\pi/4} \sec^2 x dx$.

Q15. Differentiate: $f(x) = x^2 \cos x$.

Q16. Evaluate: $\lim_{x \rightarrow 0} \frac{e^x - 1}{x}$.

8.3 Section-C (10 Marks)

Q17. Evaluate: $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2}$.

Q18. Find maxima/minima of $f(x) = x^3 - 6x^2 + 9x + 2$.

Q19. Evaluate: $\int x^2 \sin x dx$ (By Parts).

Q20. Evaluate: $\int_0^{\pi/2} \sin^2 x dx$.

8.4 Section-E (1 Mark)

Q21. What is derivative of x^n ?

Q22. What is $\int \sin x dx$?

Q23. What is $\int \frac{1}{x} dx$?

Q24. What is derivative of $\sin x$?

Q25. What is $\int e^x dx$?

Q26. What is $\lim_{x \rightarrow 0} \frac{\sin x}{x}$?

Q27. What is range of $\sin x$?

Q28. What is derivative of $\tan x$?

Q29. What is $\int \sec^2 x \, dx$?

Q30. What is chain rule?

9 Quick Revision Formulas

9.1 Functions

- $(f \circ g)(x) = f(g(x))$
- Domain of $\sqrt{x}$: $[0, \infty)$
- Range of $\sin x$: $[-1, 1]$

9.2 Limits

- $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$
- $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{1}{2}$
- $\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1$

9.3 Derivatives

- $\frac{d}{dx}(x^n) = nx^{n-1}$
- $\frac{d}{dx}(\sin x) = \cos x$
- $\frac{d}{dx}(\cos x) = -\sin x$
- $\frac{d}{dx}(e^x) = e^x$
- $\frac{d}{dx}(\ln x) = 1/x$

9.4 Integrals

- $\int x^n dx = \frac{x^{n+1}}{n+1} + C$
- $\int \sin x dx = -\cos x + C$
- $\int \cos x dx = \sin x + C$
- $\int e^x dx = e^x + C$
- $\int \frac{1}{x} dx = \ln |x| + C$

9.5 Definite Integrals

- $\int_a^b f(x) dx = F(b) - F(a)$
- If f is odd: $\int_{-a}^a f(x) dx = 0$
- If f is even: $\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx$

9.6 Maxima/Minima

- $f'(x) = 0 \rightarrow$ Critical points
- $f''(x) > 0 \rightarrow$ Minimum
- $f''(x) < 0 \rightarrow$ Maximum

Exam Tips

1. **Functions:** Domain/Range problems are easy marks
2. **Limits:** Standard limits must be memorized
3. **Derivatives:** Chain rule is most important
4. **Max/Min:** Second derivative test is quicker
5. **Integration:** Practice substitution method
6. **Definite:** Even/Odd properties save time

— End of Unit-IV Notes —